

Notes: Murphy, Shleifer, and Vishny (JPE, 1989)

Industrialization and the Big Push

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1 Big picture

- **Question.** When can simultaneous industrialization across many sectors be self-sustaining, even if no single sector can profitably industrialize on its own?
- **Setting.** The paper studies stylized **general-equilibrium models of industrialization** with **increasing returns**, **fixed costs**, **monopolistic producers**, and **aggregate demand spillovers**. The benchmark interpretation is Rosenstein–Rodan’s **big push**: a move from a bad preindustrial equilibrium to a good industrial equilibrium.
- **Core challenge.** It is not enough to say that one firm’s industrialization increases income and therefore demand for other firms. **Pecuniary externalities alone do not automatically generate multiple equilibria.** The paper asks exactly what additional conditions are needed.
- **Main theoretical result.** A big push requires a situation in which a firm’s industrialization **benefits other firms more than it benefits itself privately**. Put differently, the firm’s **private profit** must be a poor measure of its **social contribution** to market size or to the profitability of other sectors.
- **What the paper shows.**
 - A simple one-period model with only **profit spillovers** has a **unique equilibrium**; it does *not* generate a big push.
 - A big push becomes possible once industrialization affects other sectors through channels that are **not fully internalized by private profits**.
 - The paper gives **three such mechanisms**:
 1. a **factory wage premium** that raises workers’ income and manufactured-goods demand,
 2. **intertemporal investment** that shifts income from today to tomorrow,
 3. **shared infrastructure** such as a railroad, where one sector’s investment helps cover a common fixed cost.
- **Why it matters.** The paper gives a disciplined version of the old balanced-growth intuition. It explains when **coordination across sectors** can be efficient and when industrial policy may be justified, especially in economies with **small domestic markets**, **trade frictions**, or **shared infrastructure bottlenecks**.
- **Main substantive conclusions.**
 - Profit spillovers that work *only* through dividends are too weak; they do not create coexistence of industrial and nonindustrial equilibria.
 - A big push becomes possible when industrialization changes the **composition** or **timing** of demand, or lowers the **shared fixed cost** faced by other sectors.
 - In the successful big-push models, the industrial equilibrium is typically **Pareto-preferred**.
 - But multiplicity and welfare ranking are **not the same thing**: a model can have multiple equilibria without the high-investment equilibrium necessarily being better.
- **Policy takeaway.** Simultaneous industrialization can require **coordination**, **temporary subsidies**, or **public infrastructure support**. A coordinated program can achieve industrialization

at a lower explicit policy cost than piecemeal interventions because sectors implicitly subsidize one another through market-size effects.

- **Economic takeaway.** The paper’s core message is simple: a big push is possible only when private investment decisions are **too pessimistic** from a social point of view because firms ignore how their industrialization expands the market or lowers costs for other industrializing sectors.

2 Data and measurement (key objects)

2.1 This is a theory paper: what the “key objects” are

- There is **no paper-specific micro dataset**. Instead, the important objects are the **primitives of the model** and the **motivating empirical facts** on domestic-market importance.
- The paper is organized around a sequence of simple models, each asking whether an industrial equilibrium can coexist with a preindustrial one and whether the industrial equilibrium is socially preferable.

2.2 Motivating empirical facts on domestic markets

- Except for the infrastructure model, the paper’s argument relies heavily on **domestic market size**. The authors therefore begin by defending the empirical relevance of domestic demand even in an open-economy world.
- Using Chenery and Syrquin’s development evidence, the paper emphasizes that domestic demand accounted for most of industrial-output growth:
 - in countries with population above 20 million, about **72%–74%** of industrial-output growth,
 - in small primary-goods-oriented countries, about **70%–72%**,
 - in small manufacturing-oriented countries, about **50%–60%**.
- Korea, despite being a classic export-oriented example, still had about **53%** of industrial-output growth explained by domestic-demand expansion over 1955–73, and arguably more once export-intensive sectors such as textiles are stripped out.
- **Interpretation.** The domestic market is not a side issue. Even when exports matter, industry is often established in the home market first. That makes market-size arguments for industrialization empirically plausible.

2.3 Core environment shared across the models

- There is a **unit interval of sectors/goods** indexed by $q \in [0, 1]$.
- The representative consumer has labor endowment L and spends income across goods. In the one-period models, the wage is normalized to one.
- In each sector there are two technologies:

- a **competitive constant-returns-to-scale (CRS)** “cottage” technology, where one unit of labor produces one unit of output;
- an **increasing-returns-to-scale (IRS)** technology available only to a sector-specific monopolist.
- IRS production requires a **fixed labor cost** F and then produces $\alpha > 1$ units of output per unit of labor. Industrialization means paying the fixed cost and switching to the IRS technology.
- Because the competitive fringe can always supply at price one, the monopolist typically charges price one as well. The relevant issue is therefore not monopoly pricing in the usual sense, but whether there is **enough market demand** to cover the fixed cost.

2.4 Key variables and parameters

- y : aggregate income.
- Π : aggregate profits.
- n : fraction of sectors that industrialize.
- F : fixed cost of industrialization.
- α : productivity of the IRS technology.
- $a \equiv (\alpha - 1)/\alpha$: per-dollar profit margin when price is one and unit marginal cost is $1/\alpha$.
- v : disutility of factory work; in the wage-premium model it generates the required factory wage premium.
- β : time discount factor in the dynamic model.
- θ : governs intertemporal substitution.
- γ : governs within-period substitution across goods.
- R : fixed cost of infrastructure in the railroad model.
- F_1, F_2 : low and high industrial fixed costs in the infrastructure model, with $F_2 > F_1$.

2.5 Timing and equilibrium concept

- **Section III model:** one period; firms decide whether to industrialize right away.
- **Section IV model:** still one period, but industrialization raises labor income through a factory wage premium.
- **Section V model:** two periods; firms invest in period 1 and reap productivity benefits in period 2.
- **Section VI model:** two periods with a shared infrastructure project that must be built before industrialization can happen.
- The paper is fundamentally about **equilibrium multiplicity**:

- a **bad equilibrium**: cottage production / no industrialization,
- a **good equilibrium**: simultaneous industrialization across sectors.
- The paper then asks a second question beyond multiplicity: when is the industrial equilibrium **Pareto-preferred**?

3 Models and empirical specifications (all equations + interpretation)

3.1 A simple aggregate-demand-spillovers model with a unique equilibrium

The representative consumer's income satisfies

$$y = \Pi + L. \quad (1)$$

If a sector industrializes, its monopolist's profit is

$$\pi = \frac{\alpha - 1}{\alpha} y - F \equiv ay - F, \quad a \equiv \frac{\alpha - 1}{\alpha}. \quad (2)$$

If a fraction n of sectors industrialize, aggregate profits are

$$\Pi(n) = n(ay - F). \quad (3)$$

Substituting into the income identity gives aggregate income as a function of industrialization:

$$y(n) = \frac{L - nF}{1 - an}. \quad (4)$$

The effect of one more industrializing sector on aggregate income is

$$\frac{dy(n)}{dn} = \frac{\pi(n)}{1 - an}. \quad (5)$$

Interpretation.

- Equation (4) shows the standard multiplier logic: IRS sectors save labor and raise profits, and those profits are spent on other goods.
- But equation (5) is the crucial object. The sign of the spillover is exactly the sign of the investing firm's own profit.
- Therefore, if industrialization is unprofitable for one firm when the economy is unindustrialized, then that investment *reduces* aggregate income and makes industrialization even less attractive for others.
- So this model has **no big push**: either all firms industrialize or none do, but there is no coexistence of a bad and a good equilibrium caused by self-fulfilling expectations.

3.2 Model with a factory wage premium

Factory work is unpleasant, so workers employed in factories must be paid a compensating wage premium:

$$w = 1 + v > 1. \quad (6)$$

Given that factory workers must be paid this higher wage, the monopolist's profit becomes

$$\pi = y \left(1 - \frac{1+v}{\alpha} \right) - F(1+v). \quad (7)$$

For industrialization to be technologically meaningful at all, productivity gains must dominate the wage premium:

$$\alpha - 1 > v. \quad (8)$$

A no-industrialization equilibrium exists if a lone investing firm would lose money when aggregate income is just cottage-income L :

$$L \left(1 - \frac{1+v}{\alpha} \right) - F(1+v) < 0. \quad (9)$$

An all-industrialization equilibrium exists if, when everyone industrializes, profits are positive:

$$\pi = \alpha(L - F) - L(1+v) > 0. \quad (10)$$

Interpretation.

- This is the first genuine big-push model in the paper.
- The key change is that even a **money-losing factory** raises demand for manufactures because it pays **higher wages** to workers.
- So the firm's *profit* is no longer the same thing as its *contribution to aggregate demand*.
- This breaks the uniqueness result from the previous model and makes coexistence of a no-industrialization equilibrium and an industrial equilibrium possible.
- The model also produces a clear welfare result: when industrialization happens, workers get the same prices, higher labor income, and some profits, so the industrial equilibrium is Pareto-better.

3.3 Dynamic model of investment

The paper next studies a two-period environment with CES-across-time and CES-across-goods preferences:

$$U = \left[\int_0^1 x_1^\gamma(q) \, dq \right]^{\theta/\gamma} + \beta \left[\int_0^1 x_2^\gamma(q) \, dq \right]^{\theta/\gamma}. \quad (11)$$

Here:

- $1/(1 - \theta)$ is the intertemporal elasticity of substitution,

- $1/(1 - \gamma)$ is the elasticity of substitution across goods within a period.

The monopolist invests F units of labor in period 1 and can then produce with the IRS technology in period 2. The paper assumes demand is sufficiently inelastic so the monopolist still charges the fringe price one:

$$\alpha < \frac{1}{\gamma}. \quad (12)$$

If $a \equiv 1 - 1/\alpha$ denotes the per-dollar marginal profit rate and β^* the equilibrium discount factor, then the present value of profits is

$$\pi = \beta^* a y_2 - F. \quad (13)$$

In the no-industrialization equilibrium,

$$y_1 = y_2 = L, \quad (14)$$

and a representative firm does not invest if

$$\pi = \beta a L - F < 0. \quad (15)$$

In the industrial equilibrium, every sector invests, so first-period income is

$$y_1 = L - F, \quad (16)$$

while second-period income is

$$y_2 = L + \pi = L + a y_1 = \alpha L. \quad (17)$$

The equilibrium discount factor must adjust so households accept higher period-2 consumption:

$$\beta^* = \beta \left(\frac{\alpha L}{L - F} \right)^{\theta-1}. \quad (18)$$

A full-industrialization equilibrium exists if

$$(a\alpha L)\beta \left(\frac{\alpha L}{L - F} \right)^{\theta-1} - F > 0. \quad (19)$$

Interpretation.

- This model's big-push logic is intertemporal rather than intratemporal.
- A firm may lose money in present-value terms when it invests alone, yet its investment still **raises second-period income**.
- That higher future income enlarges the market for other firms that also sell in period 2.
- So industrialization shifts the **timing of demand**: it reduces demand today and raises demand tomorrow.
- The paper's most subtle point appears here: **multiplicity and welfare are distinct**. A high-investment equilibrium can exist because future cash flows support other firms, but whether it is socially better depends on the relative multipliers in the two periods.

3.4 Model of investment in infrastructure

The paper then studies a railroad as a shared infrastructure project. The key assumptions are:

- cottage technologies do *not* require the railroad,
- IRS technologies do require it,
- the railroad has a large fixed cost R and zero marginal cost,
- there are two types of sectors: a fraction n has low fixed cost F_1 , the remaining fraction $1 - n$ has high fixed cost $F_2 > F_1$.

The paper assumes low-fixed-cost firms alone still cannot support the railroad:

$$n \left(\frac{aL}{1 - an} - F_1 \right) < R. \quad (20)$$

A necessary and sufficient condition for an equilibrium with a railroad and universal industrialization is

$$a\alpha L - F_2 > R. \quad (21)$$

The railroad is socially efficient whenever

$$a\alpha L - nF_1 - (1 - n)F_2 > R. \quad (22)$$

For both industrial and nonindustrial equilibria to coexist after the railroad is built, it is enough that even a low-fixed-cost sector would not invest when expected income is only L :

$$aL - F_1 < 0. \quad (23)$$

Interpretation.

- This model introduces a second kind of externality on top of demand spillovers: an investing sector helps cover the **shared fixed cost** of infrastructure.
- The railroad can fail to be built for two separate reasons:
 1. it cannot **price-discriminate** and therefore cannot capture all the surplus it creates;
 2. it faces **extrinsic uncertainty** about whether enough sectors will industrialize and become users.
- So the market can fail even when the railroad is socially desirable and even when industrialization *could* be sustained if the railroad were somehow already in place.
- This is the strongest coordination-failure version of the paper: firms need the railroad to industrialize, but the railroad needs firms to industrialize before it can be profitable.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the paper starts with a model that fails

- The simple one-period profit-spillover model in section III is deliberately constructed to **fail**.
- That failure is essential because it clarifies the paper's identification logic: **not every pecuniary externality generates a big push**.
- If the only spillover works through dividends and private profits, then the sign of the social spillover is tied to the sign of private profitability, and multiplicity disappears.

4.2 What identifies a genuine big push in the successful models

- The paper identifies three channels through which a firm's investment can help others even when it does *not* help itself enough privately:
 1. **higher wages** paid by factories,
 2. **future cash flow** generated by current investment,
 3. **shared fixed-cost relief** through infrastructure use.
- In all three cases, the common logic is the same:
 - the firm's **private profit** is no longer the full measure of its contribution to aggregate demand or to other firms' profitability;
 - therefore, individually unprofitable investment can still make other firms more willing to industrialize.

4.3 Why domestic markets matter here

- Except for the infrastructure model, the whole argument depends on industrial firms facing a sufficiently important **domestic market**.
- If world trade were frictionless and firms could costlessly sell to the whole world, then the domestic market would not constrain adoption of increasing returns technologies.
- That is why section II is not decorative. It is the paper's empirical defense of the mechanism: the authors need domestic demand to be quantitatively relevant for industrial takeoff.

4.4 Why the industrial equilibrium is usually better — but not always

- In the wage-premium model, the industrial equilibrium is clearly better because prices are unchanged while income is higher.
- In the dynamic-investment model, the industrial equilibrium is usually better because labor is used when its contribution to aggregate income is low and saved when its multiplier is high.

- But the paper is careful not to overstate this. It shows that **multiplicity and Pareto ranking are conceptually distinct**. If multipliers are higher in the wrong period, a high-investment equilibrium may exist without being socially superior.
- This is one of the paper's most valuable conceptual clarifications: **a big push requires a coordination problem, not automatically a welfare theorem**.

4.5 What role government can play

- The paper does *not* say that government should mechanically subsidize everything.
- Instead, it says that government can be useful when it helps coordinate mutually supporting investments:
 - encouraging multiple sectors to industrialize simultaneously,
 - supporting shared infrastructure projects,
 - using temporary tariffs or subsidies more effectively than under piecemeal industrialization.
- The infrastructure model especially highlights that public support may be needed not just because infrastructure has spillovers, but because private builders may rationally fear being left with a **white elephant**.

4.6 Relation to the literature

- The paper is a formalization and partial rehabilitation of **Rosenstein–Rodan's** big-push idea and the **balanced-growth** tradition associated with Nurkse, Scitovsky, and Fleming.
- At the same time, it is careful to narrow that doctrine. The authors show that some loose statements in the older literature are too broad: **demand spillovers must have a very specific structure**.
- It also connects development theory to ideas about **pecuniary externalities, market size, increasing returns**, and later growth theory.

5 Equations in Sections

5.1 Section II: the domestic-market evidence

Read:

- The important quantitative fact is that domestic demand explains most industrial-output growth in the development evidence the paper cites.
- This is the paper's empirical motivation for taking market-size constraints seriously.
- If you reject that premise, most of the big-push logic weakens immediately.

5.2 Section III: equations (4)–(5) are the key display

Read:

- Equation (4),

$$y(n) = \frac{L - nF}{1 - an},$$

is the aggregate-income schedule.

- Equation (5),

$$\frac{dy(n)}{dn} = \frac{\pi(n)}{1 - an},$$

is the real punchline.

- It shows that the marginal sector helps others *if and only if* it is privately profitable.
- That is why the simple profit-spillover model has no big push.

5.3 Section IV: conditions (8)–(10) are the big-push region

Read:

- Condition (8), $\alpha - 1 > v$, says factory production must be productive enough to justify the wage premium technologically.
- Condition (9) says a lone factory still loses money.
- Condition (10) says all factories together make money.
- When those conditions overlap, the economy has exactly the kind of multiple equilibria Rosenstein–Rodan had in mind.

5.4 Section V: conditions (15) and (19) summarize the dynamic story

Read:

- Condition (15), $\beta aL - F < 0$, says solo investment is unattractive.
- Condition (19) says that when everyone invests, the higher future income makes investment worthwhile.
- The big insight is that investment can shift demand across time in a way that other investors value, even if the original investor does not capture enough of that benefit privately.

5.5 Section VI: conditions (20)–(23) summarize the infrastructure trap

Read:

- Condition (20) says low-fixed-cost sectors alone cannot finance the railroad.

- Condition (21) says the railroad can be sustained if everyone industrializes.
- Condition (22) says the railroad may be socially efficient even when it is privately unprofitable.
- Condition (23) shows that even after the railroad is built there can still be a bad equilibrium, so private railroad builders may rationally refuse to build.
- This is the paper's clearest statement of the coordination-failure problem.

6 Short summary

- **Method.** The paper compares a sequence of simple imperfect-competition models to isolate exactly when coordinated industrialization can be self-sustaining.
- **Identification.** A big push requires that a firm's investment increase other firms' profitability through channels **not captured by the firm's own profits**.
- **Application / interpretation.** The theory is aimed at less developed economies in which domestic markets are small, trade is costly or limited, and shared infrastructure is important.
- **Main result.** Pure profit spillovers do not generate multiplicity. Big pushes arise only when industrialization raises demand through factory wages, shifts demand into the future, or helps finance shared infrastructure. Under those conditions, coordinated policy can improve welfare and accelerate industrialization.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that **simultaneous industrialization can be self-sustaining**, but only under specific conditions.
- The necessary idea is not just "demand spillovers" in a vague sense.
- Rather, a big push requires a wedge between a firm's **private return** and its **social contribution** to the profitability of other industrializing sectors.

7.2 Economic intuition

- If a firm only helps others by paying dividends when it is already profitable, then unprofitable investment hurts the economy rather than helps it.
- But if a firm industrializes and thereby
 - pays higher wages,
 - creates future demand,
 - or helps cover a shared infrastructure cost,

then it may help other firms industrialize even if it cannot break even alone.

- That is the economic core of the big push.

7.3 Takeaway

This paper remains important because it gives the big-push idea a sharp and disciplined form. It shows why **coordination failures can matter for industrialization**, but it also narrows the circumstances in which they do. The right lesson is not that every poor country needs blanket protection or subsidies. The deeper lesson is that when industrial sectors are linked through market size, timing of demand, or shared infrastructure, **piecemeal investment can fail even though coordinated industrialization would succeed**.